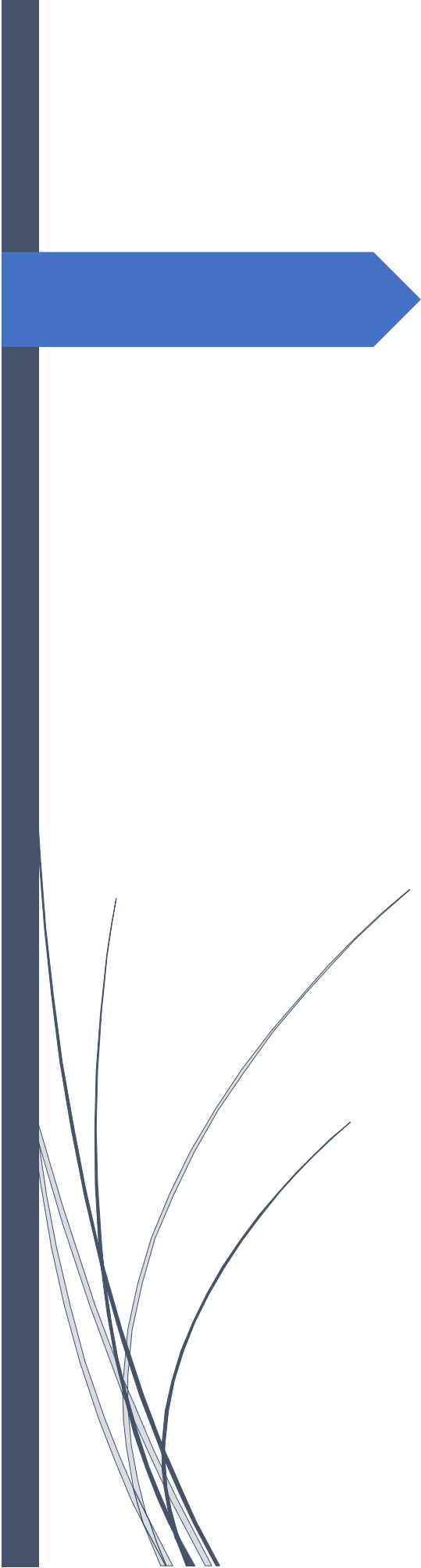




Environment , Climate change

Environmental issues: Global &
Pakistan

Environmental issues

Environmental issues in Pakistan, such as climate change, water scarcity, floods, and pollution, are not only ecological crises but also urgent civic challenges. These problems demand active community involvement and responsible citizenship to address the social, health, and economic implications they bring. Through the lens of civics and community engagement, it becomes clear that collective action, policy advocacy, and sustainable practices are essential to protect the environment and improve the quality of life for all citizens.

The most pressing environmental issues currently ongoing in Pakistan are mentioned below:

1. **Climate Change:** The list begins with the most pressing issues and the root of all issues- climate change. Even though Pakistan emissions are less than 1% of all global emissions- only **0.9%** in fact. However, it is unfortunate that Pakistan is one of the most vulnerable countries to face the negative effects of Climate Change. This will not only mean increased local and regional temperatures but also increased irregular weather patterns resulting in aggravated extreme natural events like Monsoon floods, locust attacks due to changes in habitat or because of range shifts. Climate change is responsible for heatwaves, GLOFs, floods, cyclones, and all other events that Pakistan did not suffer prior to the increase in global mean temperature.
2. **Heatwaves:** As mentioned above, heatwaves are also due to climate change and are primarily concerned with increasing temperatures and longer summers. Rainfall levels have become low in regions where summers were already quite hot and the frequency of heat waves has doubled in the past 3 years alone. Such as Multan, Jacobabad, Nawabshah, Khanpur to name a few. The 2022 heatwaves in India and Pakistan were the hottest to date and made headlines. It resulted in 90 casualties due to dehydration and heat strokes as well as unconsciousness in Pakistan in May 2022.
3. **Water Scarcity and Droughts:** Due to the low rainfall in the aftermath of heatwaves, Pakistan's Southern regions suffered water scarcity and drought due to low water tables and high evaporation rate. Especially prominent in the regions of Sibi, Tharparkar and the desert regions beyond Baluchistan. This hinders the ability to grow food greatly.
4. **Desertification:** The Extreme weather events that result in water scarcity, higher temperatures are responsible for desertification indirectly. These subsequently result in water and soil erosion due to high speed wind blowing across bare surfaces caused by deforestation for cultivation. The increasing demand for urban land and emptying rural areas mean that the grasslands or shrub lands are decreasing and eventually decreasing from many regions. Those areas or drylands which are not urbanized are overgrazed by

livestock which is also detrimental since vegetation is lost if proper livestock management plan is not there and cattle eat the grass and vegetation from the root level. This results in very slow growth of vegetation once again which is difficult in a water scarce region.

This land degradation due to urbanization, misuse of resources has surprisingly nothing to do with climate change directly per se, but is due to human activities mostly. A relatively fertile land becomes redundant and useless due to not being properly utilized.

5. **Waterlogging and Salinity:** While some regions in the south suffer from water scarcity, there are other regions which suffer from waterlogging which is what happens when water tables are filled with water and are present very close to the surface (less than 1.5m in distance from the surface). Another type of waterlogging occurs in soils that become saturated with rain water or any other runoff and the water is not removed or evaporated for a number of reasons. Salinity also results from increased salt content in water and soils that make the soil unfit for cultivation due to high sodium and salt levels.
6. **Floods:** Irregular monsoon events especially in Punjab and lower Punjab regions have become a cause of major concern due to the risk of flooding. As seen in the devastating Monsoon season induced floods of 2022. The lack of proper evacuation plans, communication, disaster management teams and overall incapability of dealing with the risk has made Pakistan particularly even vulnerable. Coupled with the fact that there is not enough evacuation tools and post disaster equipment like sump pumps for water removal and basic necessities for survivors. It is alarming how Pakistan has no initiatives to deal with floods.
7. **GLOFs:** Glacial Lake Outburst Floods. These are floods that result due to glacial ice and snow melting. Pakistan has the highest number of glaciers outside the poles. In the regions of Hindu Kush, Karakorum and Himalayas, in the Northern regions of Pakistan where temperatures used to be cooler than it is now. There are a total of 3044 Glacial lakes in Pakistan specifically in KPK and Gilgit Baltistan and all are undergoing various stages of melting rapidly. More transboundary air pollution deposits particulate matter and other pollutants on the tips or surfaces of these Glaciers which upon melting cannot be used as potable water further contributing to water scarcity in these areas. Whereby the mountain communities could use this glacial melt for drinking and cooking purposes now even that is dangerous.
8. **Air Pollution:** Perhaps the biggest hazard in Pakistan is that of pollution. Pollution of all types, i.e. air, water and land are at immediate risk of being severely degraded. Air pollution is being contributed largely by increasing number of vehicles and transportation owned by an ever-increasing population in small areas unfit for housing. This leads to poor air quality of air and increased occurrences of SMOG twice a year during the transitional season. Smog is a type of air pollution caused by a combination of smoke, fog, and

chemical fumes. Smog has been dubbed as the fifth season of Pakistan and we can glean how much of a serious problem this is for people with allergies, respiratory ailments and breathing problems who have to live through a couple of months breathing in polluted air.

9. **Water Pollution:** due to improper construction of sewage and drain systems as well as misuse of pesticides by farmers and general practice of industrial dumping of effluent in to the water ways and channels, water pollution is a major problem. It has led to the annual occurrence of a '**dengue season**' in Pakistan because people living near these water ways or using it as drinking water become regularly infected. Huge numbers of people die annually and it is a recurring epidemic in Pakistan. However, this 'epidemic' can be solved as it is man-made due to lack of investment and water sanitation initiatives. Stagnant waters across the country is a problem mainly because of blocked sewages and drains that have not been cleaned or monitored in ages. Other water borne diseases also occur like Typhoid, cholera, dysentery, sleeping sickness/Trypanosoma etc. Animals are also suffering from water pollution due to nets, fishing rods and other waste dumped in seas and rivers.
10. **Loss of Indus River Delta:** Whereas dams are necessary for flood control in Pakistan however they pose another crucial problem which is the shrinking of the Indus Delta. As the Tarbela dam and another dam were created on important tributaries of the Indus river, this has led to decreased in the delta area. The delta area that was created due to the fast flowing river waters as they emptied into the Arabian sea off the coast of Karachi, the mismanagement of water by the government as well as problems of siltation and salinity which effects its freshwater ecosystem.

Conclusion:

Addressing environmental issues is not solely the duty of the government—it is a civic responsibility that requires active engagement from individuals, communities, and institutions. Informed and committed citizens can play a powerful role by participating in awareness campaigns, supporting green policies, conserving resources, and holding stakeholders accountable. By fostering a culture of environmental stewardship through civic education and community action, we can pave the way toward a cleaner, safer, and more resilient Pakistan.

Reference:

- “Global Climate Risk Index 2021”. Germanwatch, (2021) pp, 11-14
- “Economic Survey of Pakistan (2019-20)” Government of Pakistan, Finance Division, pp. 17.
- Iberdrola (2021). “Top ten most affected countries by climate change.” <https://www.iberdrola.com/sustainability/top-ten-most-affected-countries-by-climate-change>.
- www.pulitzercenter.or.Abbas et al.,
2018Z. Abbas, H.W.T. Mapoma, C. Su, S.Z. Aziz, Y. Ma, N. Abbas
- Spatial analysis of groundwater suitability for drinking and irrigation in Lahore, Pakistan